

Lec # 05
ES111

Probability of
Event,
Additive Rules

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Discussion Plan

1. Probability of Event

1. Additive Rules

3. Practice Problems (Sec. 2.3, 2.4)



Game of chance

Game of chance



"The fix is on. Put everything on the tortoise."



I don't like to take chances.

Mathematics of Probability



Pascal



Bernoulli



Leibniz

Probability of an event

Probability of an event

1. It might rain tomorrow.

Probability of an event

1. It **might** rain tomorrow.
2. I **may** get A grade in ES-111.

Probability of an event

1. It **might** rain tomorrow.
2. I **may** get A grade in ES-111.
3. The sun **probably** rise from east.

Probability of an event

1. It **might** rain tomorrow.
2. I **may** get A grade in ES-111.
3. The sun **probably** rise from east.
4. The ES-111 class students **perhaps** get married in 2025.

Probability of an event

1. It **might** rain tomorrow. $\rightarrow 0.1$
2. I **may** get A grade in ES-111. $\rightarrow 0.5$
3. The sun **probably** rise from east. $\rightarrow 1$
4. The ES-111 class students **perhaps** get married in 2025. $\rightarrow 0$

Probability of an event

1. It **might** rain tomorrow. $\rightarrow 0.1$
2. I **may** get A grade in ES-111. $\rightarrow 0.5$
3. The sun **probably** rise from east. $\rightarrow 1$
4. The ES-111 class students **perhaps** get married in 2025. $\rightarrow 0$

Probability numerical value tells degree of certainty

Probability: Assigning weights to sample points

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Ex A coin is tossed. What is the probability of head.

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$$S = \{H, T\}$$

Probability: Assigning weights to sample points

Ex A coin is tossed. What is the probability of head.

$$S = \{H, T\} \rightarrow P(H) = \frac{1}{2}$$

Probability: Assigning weights to sample points

Ex A coin is tossed. What is The probability of head.

$$S = \{H, T\} \rightarrow P(H) = \frac{1}{2} \quad \text{But why?}$$

Probability: Assigning weights to sample points

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When probability of any event A is calculated, we assume following

Probability: Assigning weights to sample points

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When probability of any event A is calculated, we assume following

1. Probability of event A , sum of all weights of sample points in A

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$$S = \{HH, HT, TH, TT\}$$

Probability: Assigning weights to sample points

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Ex A coin is tossed twice. What is the probability that at least 1 head occurs?

$$S = \{HH, HT, TH, TT\}$$

(i) assign weight to sample points

Probability: Assigning weights to sample points

1. Probability of event A , sum of all weights of sample points in A

Ex A coin is tossed twice. What is the probability that at least 1 head occurs?

$$S = \{ \overset{\uparrow \omega}{HH}, \overset{\uparrow \omega}{HT}, \overset{\uparrow \omega}{TH}, \overset{\uparrow \omega}{TT} \}$$

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(i) assign weight to sample points

(ii) sum of all weights must be 1

Probability: Assigning weights to sample points

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Ex A coin is tossed twice. What is the probability that at least 1 head occurs?

$$S = \left\{ \overset{\uparrow \omega}{HH}, \overset{\uparrow \omega}{HT}, \overset{\uparrow \omega}{TH}, \overset{\uparrow \omega}{TT} \right\}$$

(i) assign weight to sample points

(ii) sum of all weights must be 1 $\omega + \omega + \omega + \omega = 1$

Probability: Assigning weights to sample points

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(i) assign weight to sample points

(ii) sum of all weights must be 1 $\omega + \omega + \omega + \omega = 1 \Rightarrow \omega = \frac{1}{4}$

Probability: Assigning weights to sample points

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(i) assign weight to sample points

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(iii) Identify A , and sum all weights of A

Probability: Assigning weights to sample points

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(i) assign weight to sample points

(ii) sum of all weights must be 1 $\omega + \omega + \omega + \omega = 1 \Rightarrow \omega = \frac{1}{4}$

(iii) Identify A , and sum all weights of $A \rightarrow P(A) = \frac{1}{4} + \frac{1}{4} + \frac{1}{4}$

Probability: Assigning weights to sample points

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Probability: Assigning weights to sample points

1. Probability of event A , sum of all weights of sample points in A

Ex A **die is loaded** in such a way that an **even number is twice** as likely to occur as an **odd number**. If **E is the event that a number less than 4 occurs** on a single toss of the die, find $P(E)$?

Probability: Assigning weights to sample points

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Ex A die is loaded in such a way that an even number is twice as likely to occur as an odd number. If E is the event that a number less than 4 occurs on a single toss of the die, find $P(E)$?

$$S = \{1, 2, 3, 4, 5, 6\}$$

(i) assign weight to sample points

(ii) sum of all weights must be 1

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$$S = \{ \underset{\substack{\downarrow \\ w}}{1}, \underset{\substack{\downarrow \\ w}}{2}, \underset{\substack{\downarrow \\ w}}{3}, 4, 5, 6 \}$$

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$\downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow$
 $w \quad 2w \quad w \quad 2w \quad w \quad 2w$

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$\downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow$
 $w \quad 2w \quad w \quad 2w \quad w \quad 2w$

(i) assign weight to sample points $\rightarrow 9w$

(ii) sum of all weights must be 1

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(i) assign weight to sample points $\rightarrow 9w$

(ii) sum of all weights must be 1 $\rightarrow 9w = 1$

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 $w \quad 2w \quad w \quad 2w \quad w \quad 2w$

(i) assign weight to sample points $\rightarrow 9w$

(ii) sum of all weights must be 1 $\rightarrow 9w = 1 \rightarrow w = \frac{1}{9}$

(iii) Identify E , and sum all weights of E

Probability: Assigning weights to sample points

1. Probability of event A , sum of all weights of sample points in A

Ex A die is loaded in such a way that an even number is twice as likely to occur as an odd number. If E is the event that a number less than 4 occurs on a single toss of the die, find $P(E)$?

$$S = \{ \boxed{1, 2, 3, 4}, 5, 6 \}$$

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 $w \quad 2w \quad w \quad 2w \quad w \quad 2w$

(i) assign weight to sample points $\rightarrow 9w$

(ii) sum of all weights must be 1 $\rightarrow 9w = 1 \rightarrow w = \frac{1}{9}$

(iii) Identify E , and sum all weights of E $P(E) = w + 2w + w + 2w$

Probability: Assigning weights to sample points

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(i) assign weight to sample points $\rightarrow 9w$

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(iii) Identify E , and sum all weights of E $P(E) = \frac{1}{9} + 2w + \frac{1}{9} + 2w$

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(iii) Identify E , and sum all weights of E $P(E) = \frac{1}{9} + \frac{2}{9} + \frac{1}{9} + \frac{2}{9}$ Ans



A die is loaded in such a way that an even number is twice as likely to occur as an odd number. let A be the event that an even number turns up and let B be the event that a number divisible by 3 occurs. Find $P(A \cup B)$ and $P(A \cap B)$.



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$$S = \{1, 2, 3, 4, 5, 6\}$$

from previous question

$$P(\text{odd}) = \frac{1}{9}$$



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from previous question

$$P(\text{odd}) = \frac{1}{9}$$

$$A = \{2, 4, 6\}$$

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$$B = \{3, 6\}$$

$$A \cup B = \{2, 3, 4, 6\}$$



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$$A \cup B = \{2, 3, 4, 6\}$$

$$A \cap B = \{6\}$$



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$$S = \{1, 2, 3, 4, 5, 6\}$$

from previous question

$$P(\text{odd}) = \frac{1}{9}$$

$$P(\text{even}) = \frac{2}{9}$$

$$A = \{2, 4, 6\}$$

$$B = \{3, 6\}$$

$$A \cup B = \{\cancel{2}, 3, \cancel{4}, \cancel{6}\} \rightarrow P(A \cup B) = \frac{2}{9} + \frac{2}{9} + \frac{2}{9}$$

$$A \cap B = \{6\}$$



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from previous question

$$P(\text{odd}) = \frac{1}{9}$$

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$$A = \{2, 4, 6\}$$

$$B = \{3, 6\}$$

$$A \cup B = \{\cancel{2}, \cancel{3}, \cancel{4}, \cancel{6}\} \rightarrow P(A \cup B) = \frac{2}{9} + \frac{1}{9} + \frac{2}{9} + \frac{2}{9} = \frac{7}{9} \text{ Ans}$$

$$A \cap B = \{6\}$$



A die is loaded in such a way that an even number is twice as likely to occur as an odd number. let A be the event that an even number turns up and let B be the event that a number divisible by 3 occurs. Find $P(A \cup B)$ and $P(A \cap B)$.

$$S = \{1, 2, 3, 4, 5, 6\}$$

from previous question

$$P(\text{odd}) = \frac{1}{9}$$

$$A = \{2, 4, 6\}$$

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$$B = \{3, 6\}$$

$$A \cup B = \{\cancel{2}, \cancel{3}, \cancel{4}, \cancel{6}\} \rightarrow P(A \cup B) = \frac{2}{9} + \frac{1}{9} + \frac{2}{9} + \frac{2}{9} = \frac{7}{9} \text{ Ans}$$

$$A \cap B = \{\cancel{6}\} \rightarrow P(A \cap B) = \frac{2}{9} \text{ Ans}$$

Probability for equally likely events

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Probability of A =

Probability for equally likely events

Probability of A = number of favorable outcomes to A

Probability for equally likely events

$$\text{Probability of } A = \frac{\text{number of favorable outcomes to } A}{\text{Total number of possible outcomes}}$$

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$$P(A) =$$

Probability for equally likely events

Probability of A = $\frac{\text{number of favorable outcomes to } A}{\text{Total number of possible outcomes}}$

$$P(A) = \frac{n}{N}$$

Probability for equally likely events

Probability of A = $\frac{\text{number of favorable outcomes to } A}{\text{Total number of possible outcomes}}$

$$P(A) = \frac{n}{N}$$

e.g.

$$S = \{1, 2, 3, 4, 5, 6\}$$

Probability for equally likely events

Probability of A = $\frac{\text{number of favorable outcomes to } A}{\text{Total number of possible outcomes}}$

$$P(A) = \frac{n}{N}$$

e.g. $S = \{1, 2, 3, 4, 5, 6\}$

Probability of even numbers

Probability for equally likely events

Probability of A = $\frac{\text{number of favorable outcomes to } A}{\text{Total number of possible outcomes}}$

$$P(A) = \frac{n}{N}$$

e.g. $S = \{1, 2, 3, 4, 5, 6\}$

Probability of even numbers $\rightarrow P(E) =$

Probability for equally likely events

Probability of A = $\frac{\text{number of favorable outcomes to } A}{\text{Total number of possible outcomes}}$

$$P(A) = \frac{n}{N}$$

→ favorable

e.g

$$S = \{1, 2, 3, 4, 5, 6\}$$

Probability of even numbers → $P(E) =$

Probability for equally likely events

Probability of A = $\frac{\text{number of favorable outcomes to } A}{\text{Total number of possible outcomes}}$

$$P(A) = \frac{n}{N}$$

→ favorable

e.g

$$S = \{1, 2, 3, 4, 5, 6\}$$

Probability of even numbers → $P(E) = \underline{3}$

Probability for equally likely events

Probability of A = $\frac{\text{number of favorable outcomes to } A}{\text{Total number of possible outcomes}}$

$$P(A) = \frac{n}{N}$$

→ favorable

e.g

$$S = \{1, 2, 3, 4, 5, 6\}$$

Probability of even numbers → $P(E) = \frac{3}{6}$
Total outcomes



A statistics class for engineers consists of 25 industrial, 10 mechanical, 10 electrical, and 8 civil engineering students. If a person is randomly selected by the instructor to answer a question, find the probability that the student chosen is (a) an industrial engineering major and (b) a civil engineering or an electrical engineering major.



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Ⓐ $P(I) = \frac{25}{53}$



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Ⓐ

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Union



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(a)

$$P(I) = \frac{25}{53}$$

u

(b)

$$P(c \cup e) = \frac{18}{53} \quad \underline{\underline{Ans}}$$

Additive Rules of Probability

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Let A and B are two events

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Let A and B are two events, we want to add the probabilities

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Let A and B are two events, we want to add the probabilities

$$P(A \cup B) = ?$$

Additive Rules of Probability

Let A and B are two events, we want to add the probabilities

$$P(A \cup B) = P(A) + P(B)$$

Additive Rules of Probability

Let A and B are two events, we want to add the probabilities

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Additive Rules of Probability

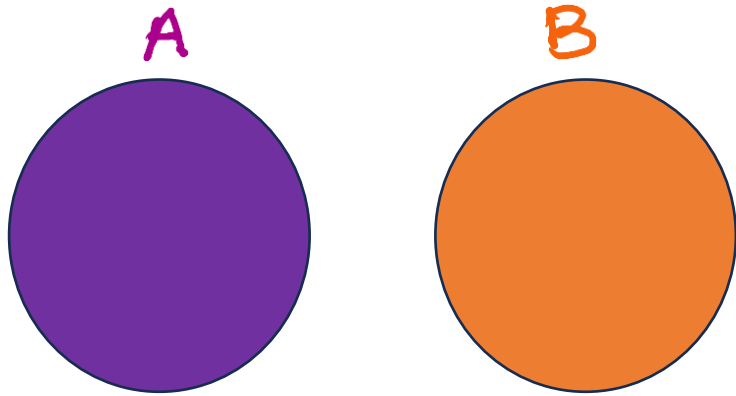
Let A and B are two events, we want to add the probabilities

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) \rightarrow \text{why?}$$

Additive Rules of Probability

Let A and B are two events, we want to add the probabilities

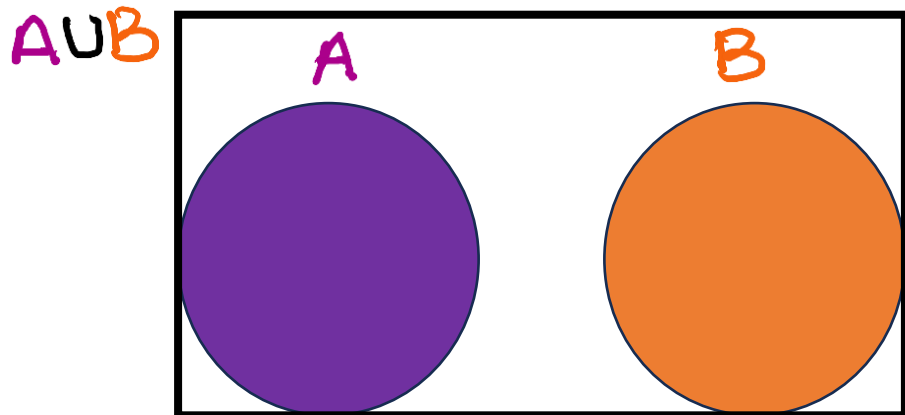
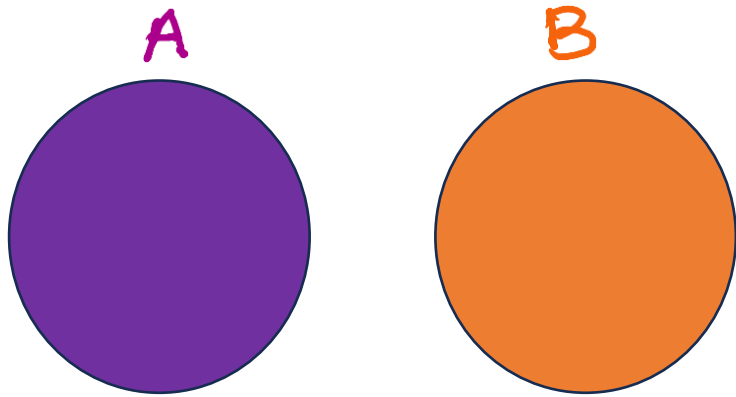
$$P(A \cup B) = P(A) + P(B) - \boxed{P(A \cap B)} \rightarrow \text{why?}$$



Additive Rules of Probability

Let A and B are two events, we want to add the probabilities

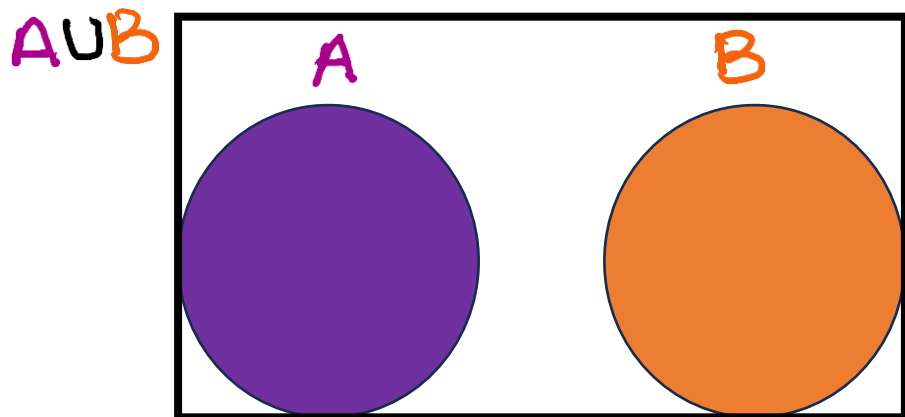
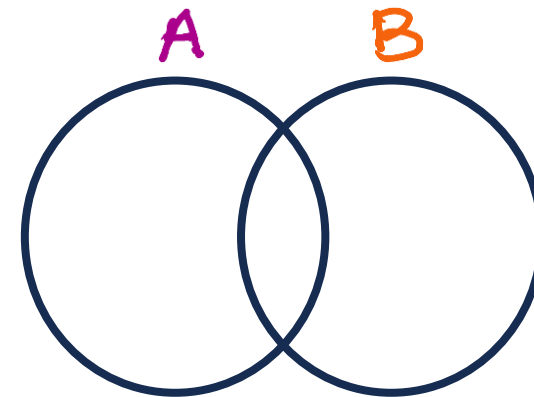
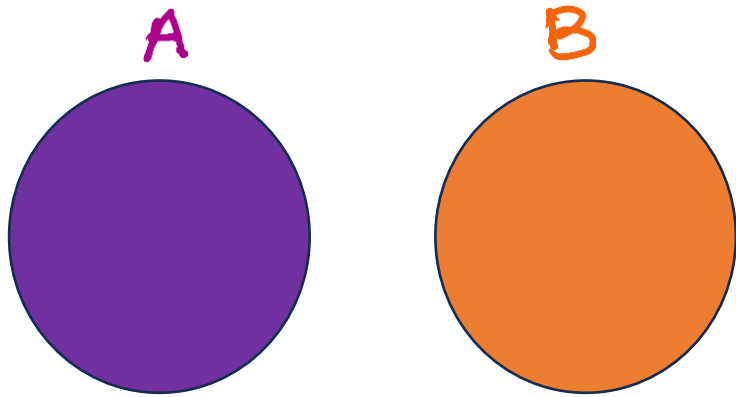
$$P(A \cup B) = P(A) + P(B) - \boxed{P(A \cap B)} \rightarrow \text{why?}$$



Additive Rules of Probability

Let A and B are two events, we want to add the probabilities

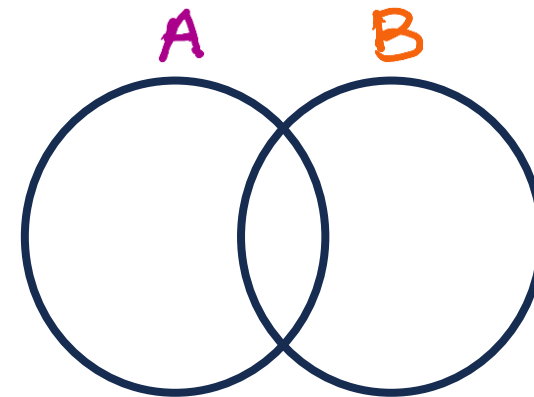
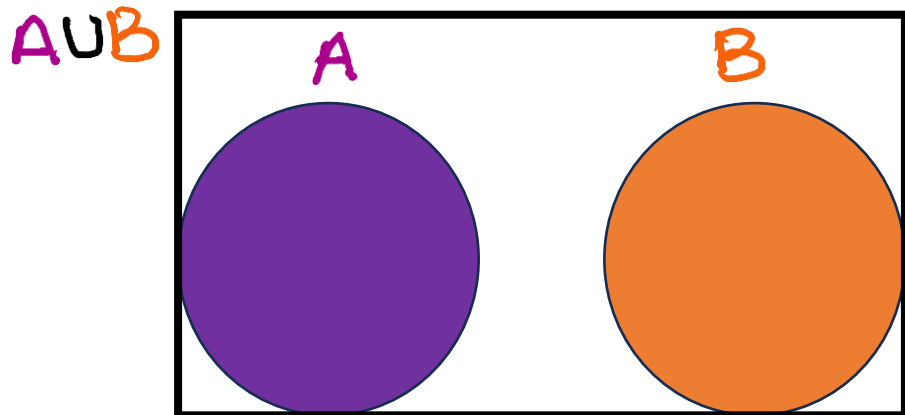
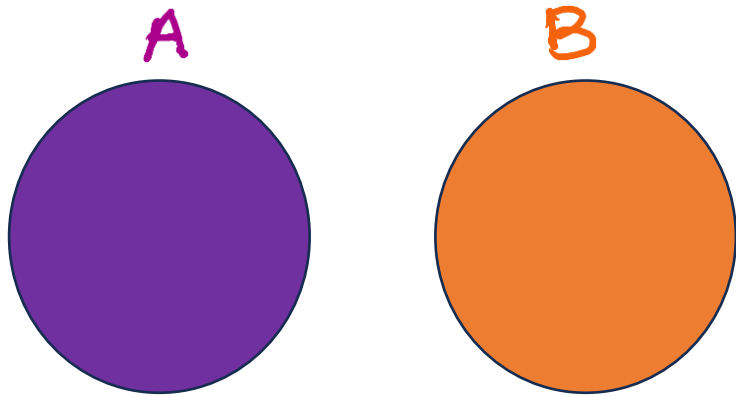
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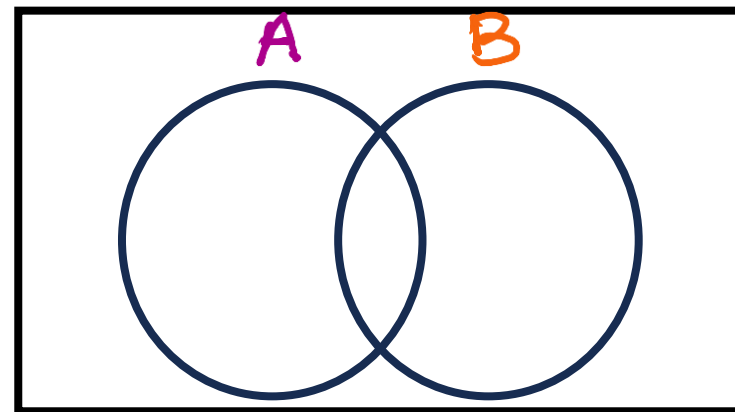
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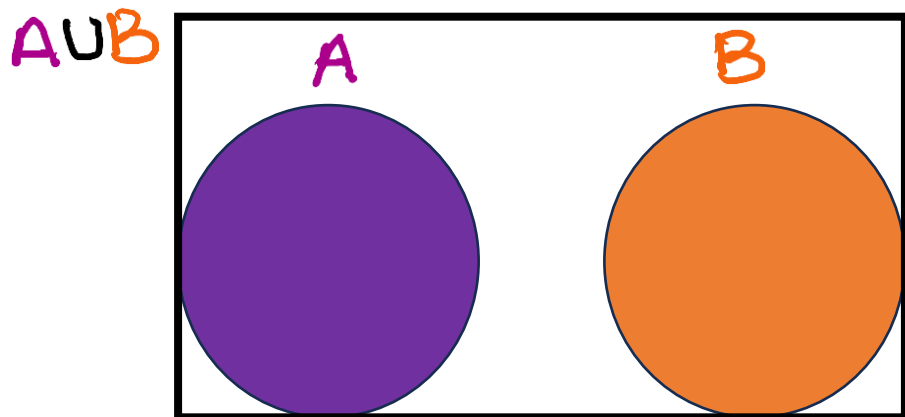
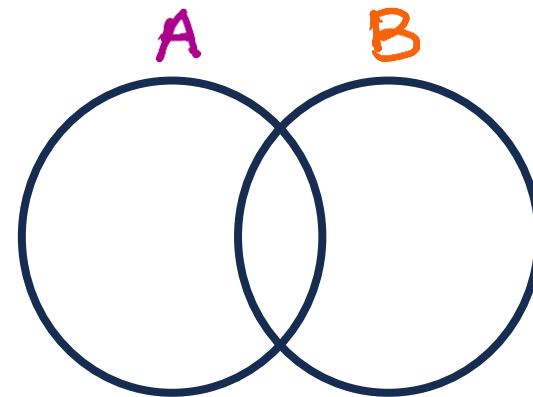
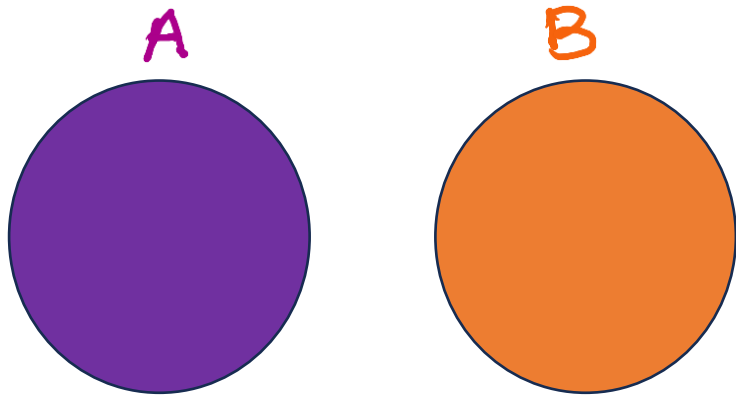
$A \cup B$



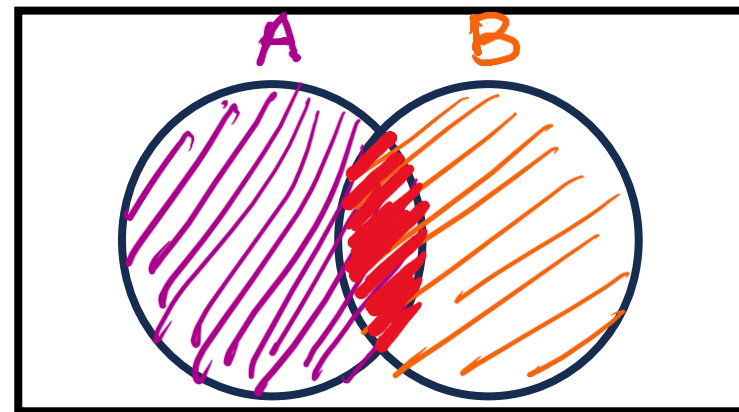
Additive Rules of Probability

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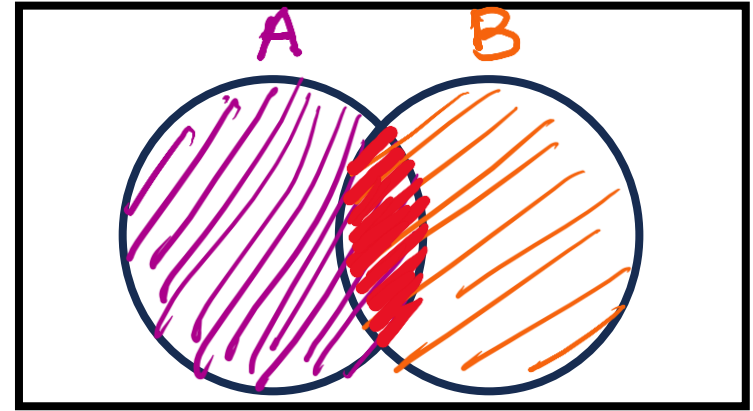


$$\begin{aligned} &A \cup B \\ &P(A) + \\ &P(B) + \\ &- P(A \cap B) \end{aligned}$$



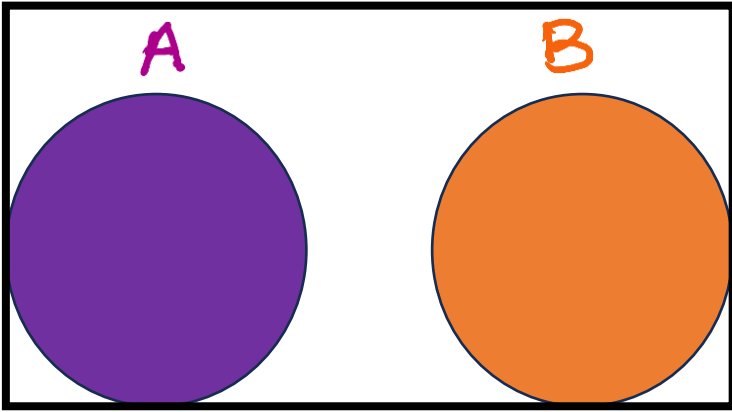
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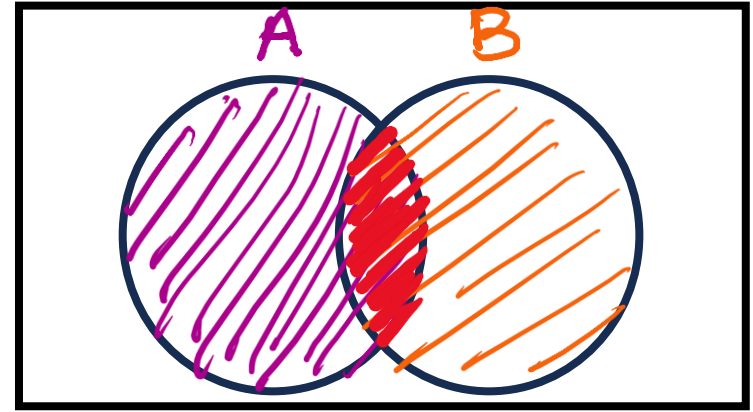


Additive Rules of Probability

$A \cup B$

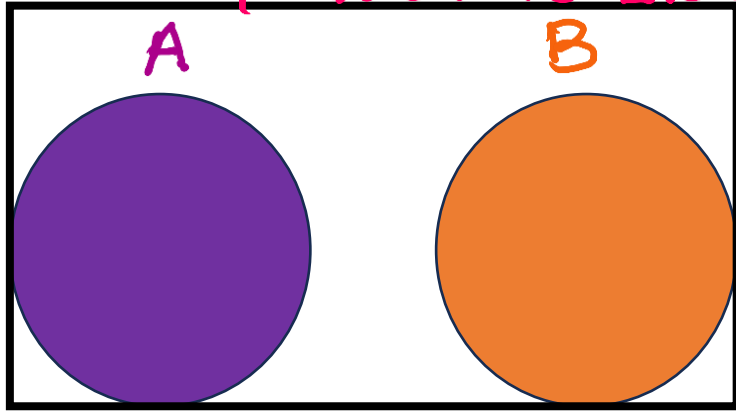


$$A \cap B = \{\emptyset\}$$

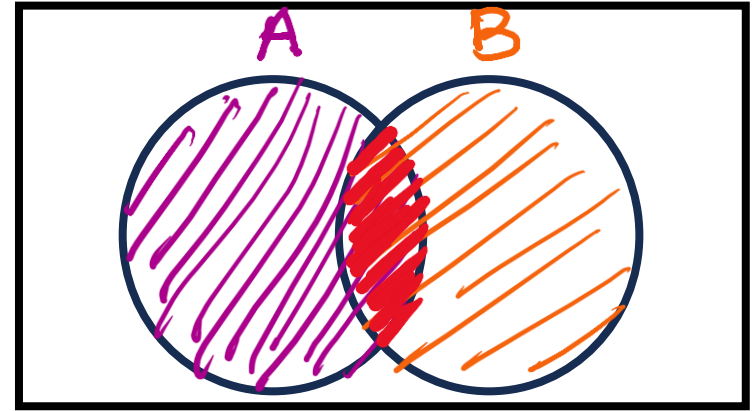


Additive Rules of Probability

$A \cup B$ Mutually Exclusive Events

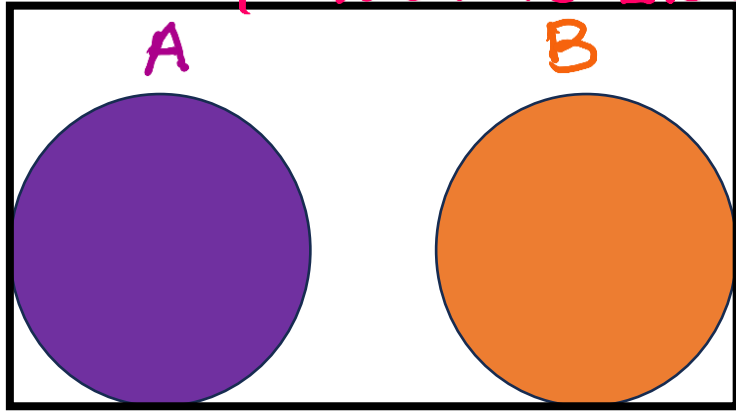


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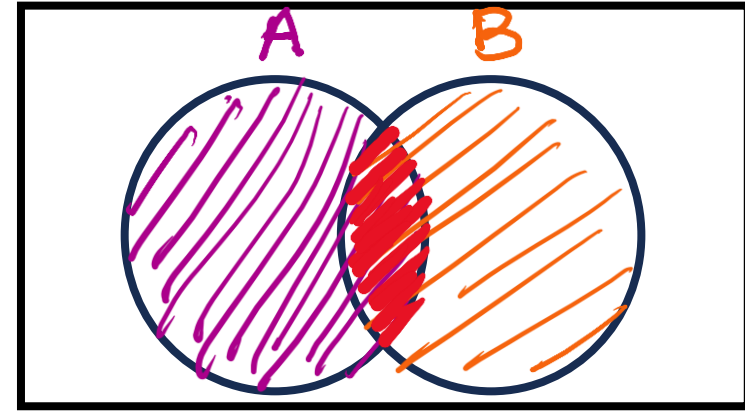


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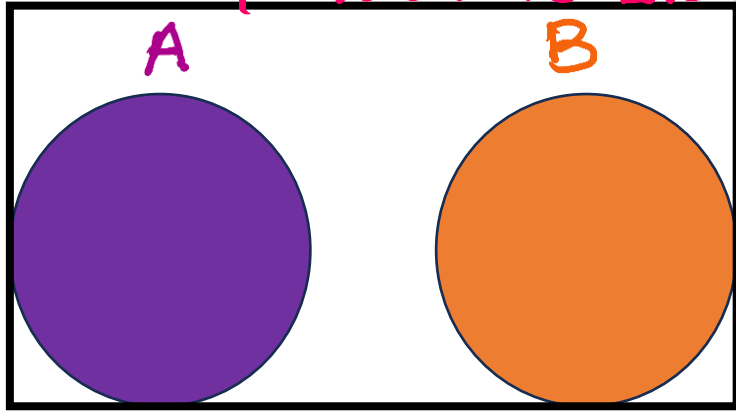
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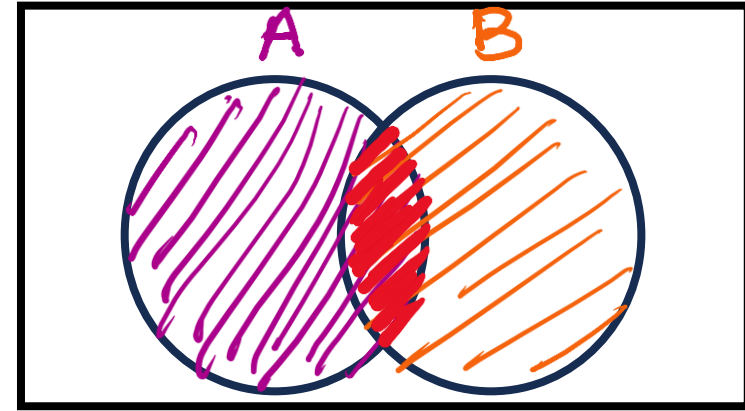
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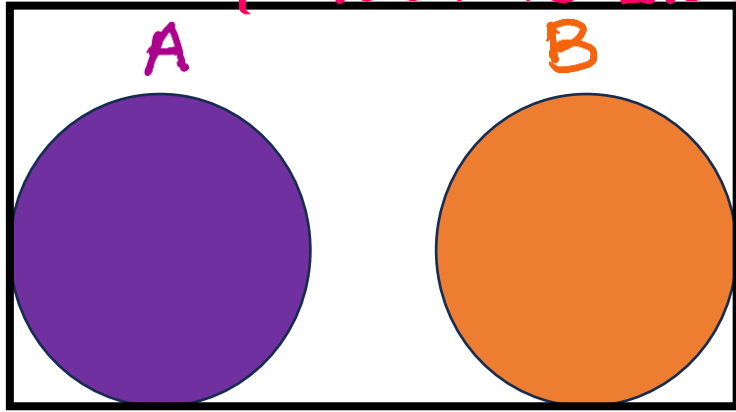


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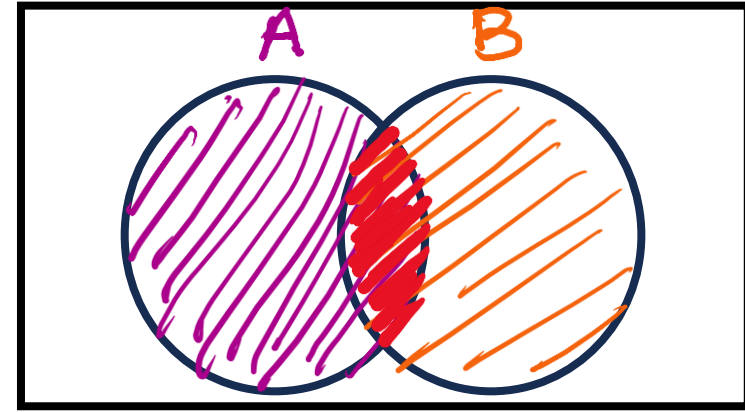
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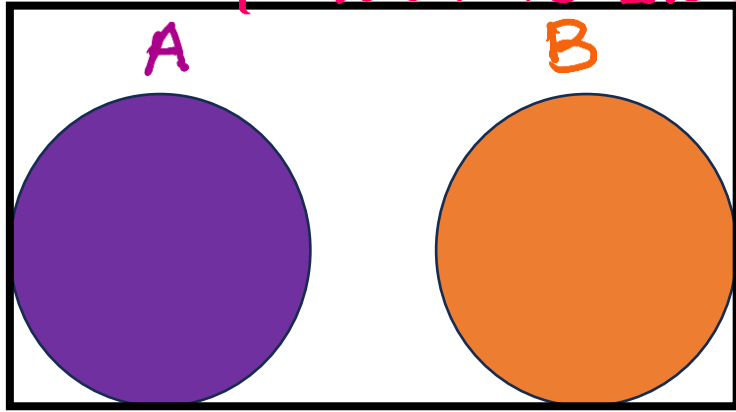


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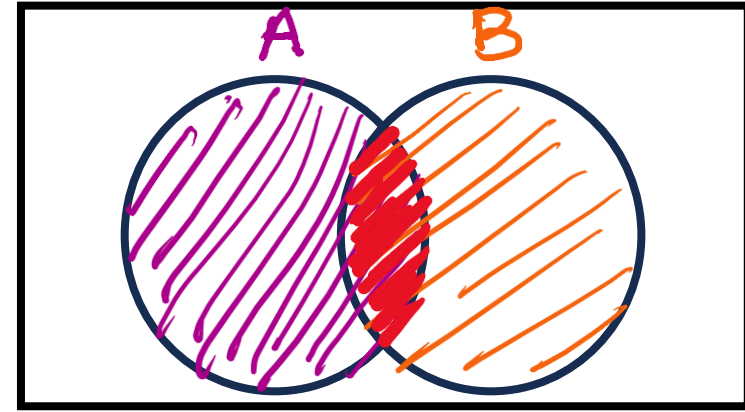


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$$P(A \cup B) = P(A) + P(B)$$

↙

$$P(A_1 \cup A_2 \cup A_3 \dots \cup A_n) =$$

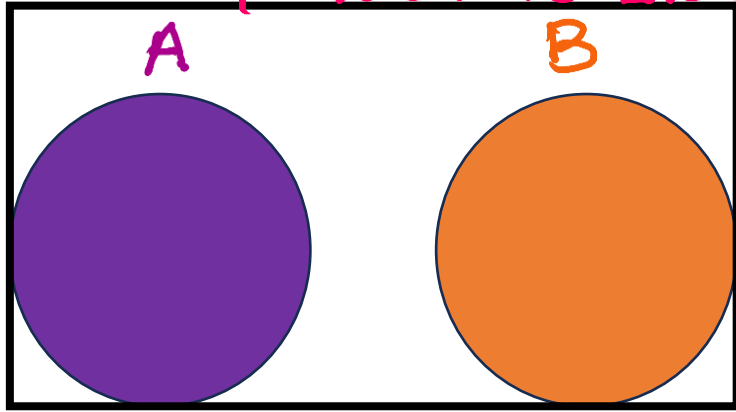


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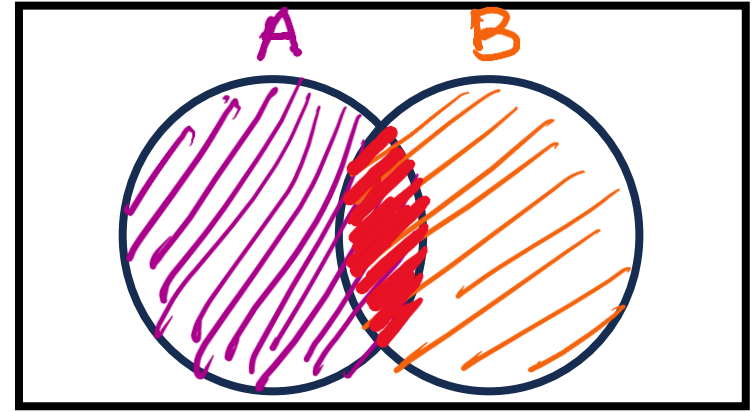
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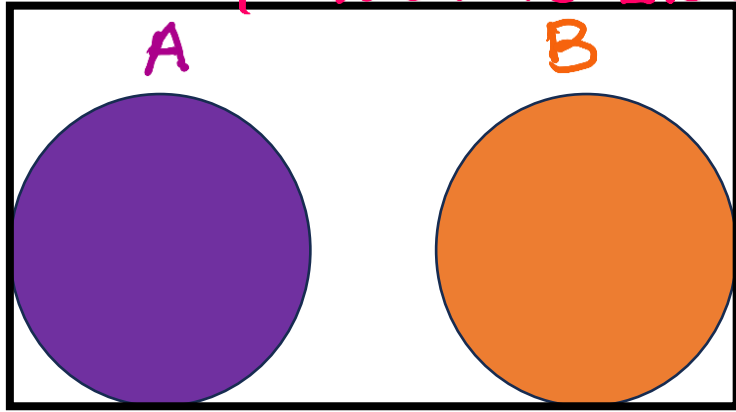
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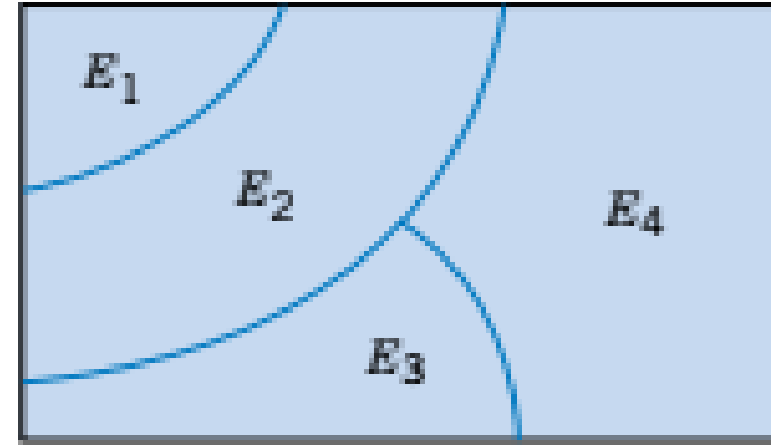
Additive Rules of Probability

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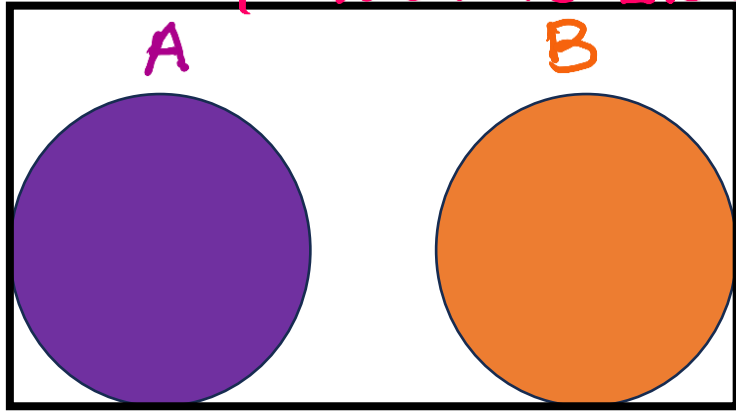
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Mutually Exclusive Events



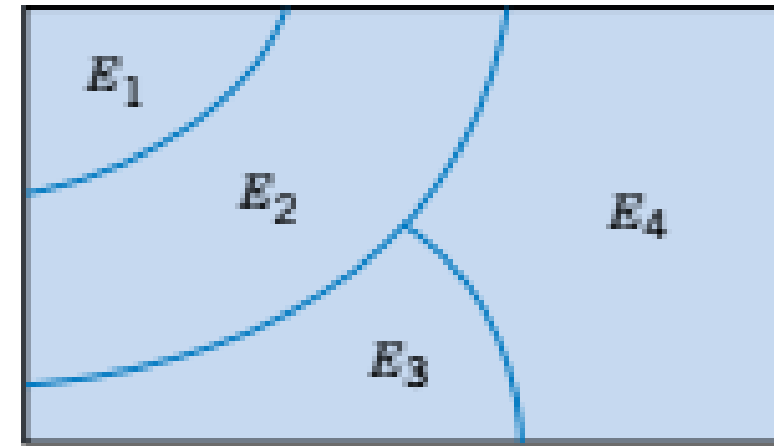
Additive Rules of Probability

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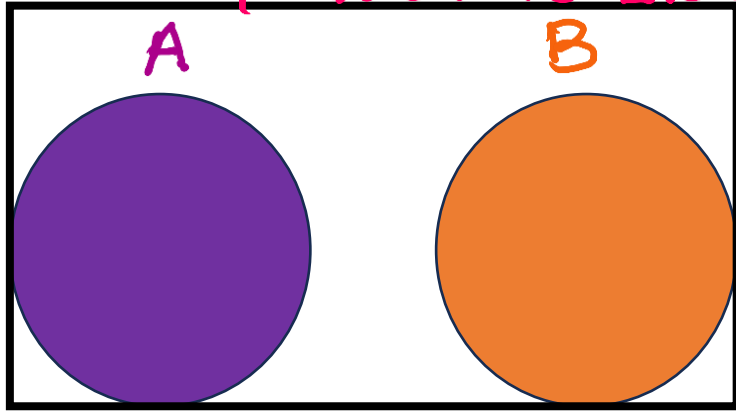
Mutually Exclusive Events



$$P(E_1) + P(E_2) + P(E_3) + P(E_4)$$

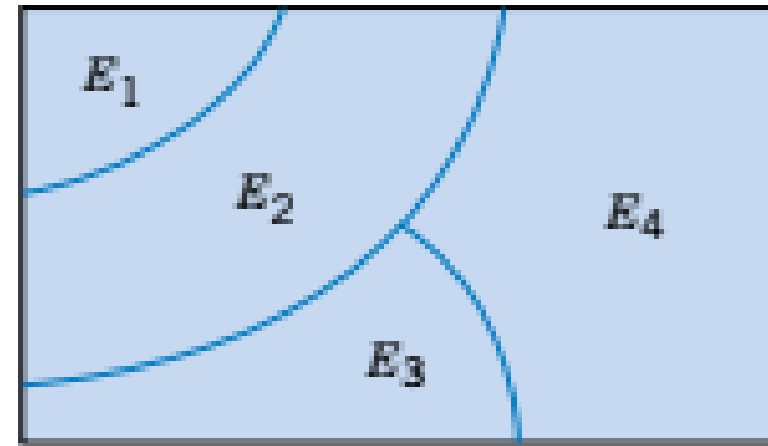
Additive Rules of Probability

$A \cup B$ Mutually Exclusive Events



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Mutually Exclusive Events



$$P(E_1) + P(E_2) + P(E_3) + P(E_4) = P(S)$$

Sample space



John is going to graduate from an industrial engineering department in a university by the end of the semester. After being interviewed at two companies he likes, he assesses that his probability of getting an offer from company A is 0.8, and his probability of getting an offer from company B is 0.6. If he believes that the probability that he will get offers from both companies is 0.5, what is the probability that he will get at least one offer from these two companies?



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$$\begin{aligned} P(A \cup B) &= P(A) + P(B) - P(A \cap B) \\ &= 0.8 + 0.6 - 0.5 \end{aligned}$$



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$$= 0.8 + 0.6 - 0.5 = 0.9 \text{ Ans}$$



What is the probability of getting a total of 7 or 11 when a pair of fair dice is tossed?



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A

B



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SAMPLE SPACE FOR
A PAIR OF DICE

	1	2	3	4	5	6
1	2	3	4	5	6	7
2	3	4	5	6	7	8
3	4	5	6	7	8	9
4	5	6	7	8	9	10
5	6	7	8	9	10	11
6	7	8	9	10	11	12

6

2



What is the probability of getting a total of 7 or 11 when a pair of fair dice is tossed?

A

B

$$P(A \cup B) = P(A) + P(B)$$

$$= \frac{6}{36} + \frac{2}{36} \quad \text{Ans}$$

$$A \cap B = \{\emptyset\}$$

SAMPLE SPACE FOR
A PAIR OF DICE

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6	7	8	9	10	11	12

6

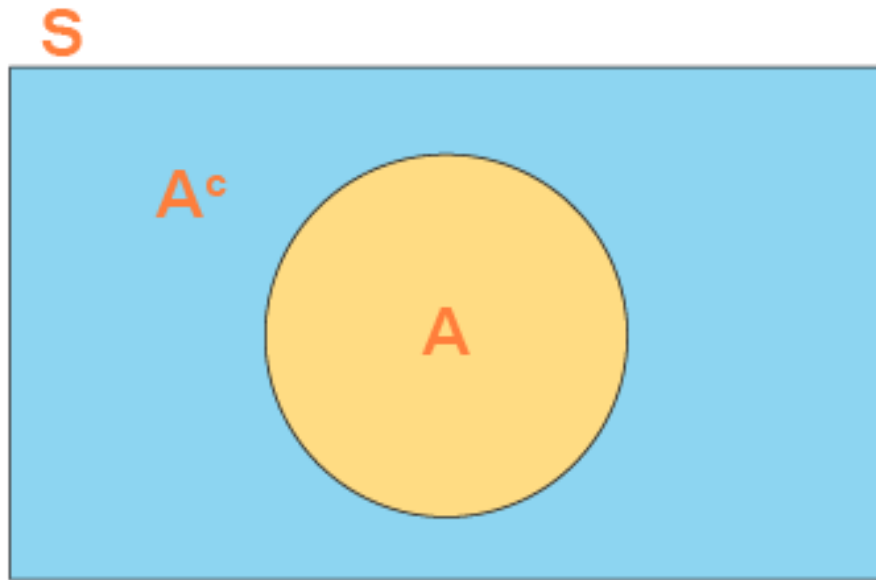
2

Additive Rules of Probability

If A and A' are complementary events

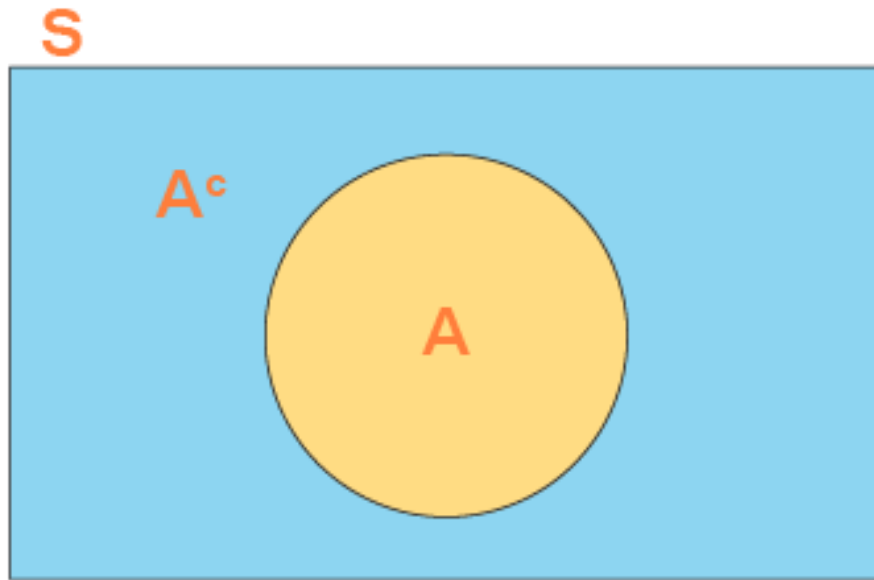
Additive Rules of Probability

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Additive Rules of Probability

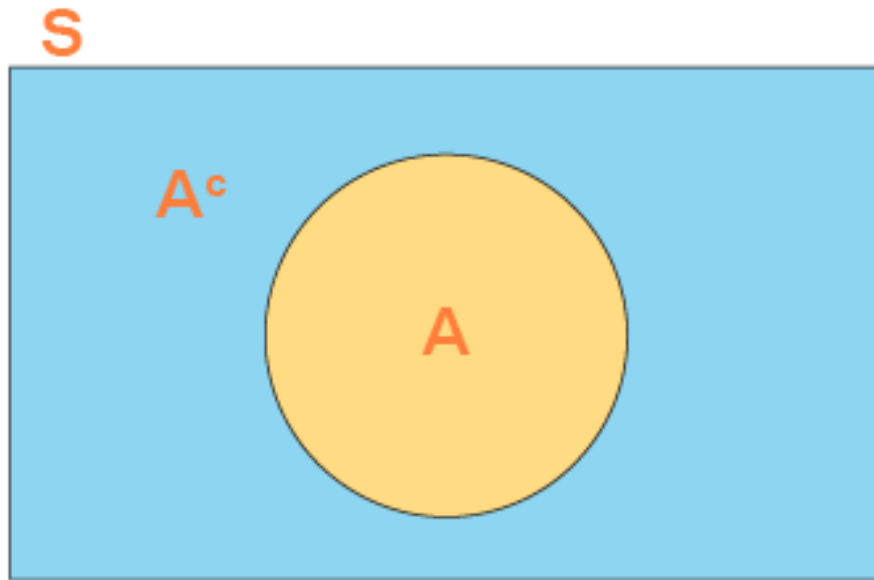
If A and A' are complementary events



$$P(A) + P(A') =$$

Additive Rules of Probability

If A and A' are complementary events



$$P(A) + P(A') = P(S)$$



If the probabilities that an automobile mechanic will service 3, 4, 5, 6, 7, or 8 or more cars on any given workday are, respectively, 0.12, 0.19, 0.28, 0.24, 0.10, and 0.07, what is the probability that he will service at least 5 cars on his next day at work?



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Let *probability* that he service at least **5** cars $P(E)$



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← unknown



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$$P(E) + P(E') = P(S)$$

$$P(E) + P(E') = 1$$



If the probabilities that an automobile mechanic will service 3, 4, 5, 6, 7, or 8 or more cars on any given workday are, respectively, 0.12, 0.19, 0.28, 0.24, 0.10, and 0.07, what is the **probability that he will service at least 5 cars** on his next day at work?

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$$= 1 -$$



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$$= 1 - 0.31$$



If the probabilities that an automobile mechanic will service 3, 4, 5, 6, 7, or 8 or more cars on any given workday are, respectively, 0.12, 0.19, 0.28, 0.24, 0.10, and 0.07, what is the probability that he will service at least 5 cars on his next day at work?

Let probability that he service at least 5 cars $P(E)$

unknown

$$P(E) + P(E') = P(S)$$

$$P(E) + P(E') = 1$$

$$P(E) = 1 - P(E')$$

$$= 1 - 0.31 = 0.69 \text{ Ans}$$